

# Corporate Finance

Lecture 3 extras: Read the news, and look around you

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Extra 1

**Read the news: Eni**

## Eni: a big dividend payer in Milan



Palazzo Eni, Rome

- ▶ Italian oil and gas group, listed in Milan (FTSE MIB) and New York. The state holds about 30%.
- ▶ Share price about **EUR 24**, market cap about **EUR 60 billion**.
- ▶ 2025: adjusted net profit **EUR 5.0 billion**, operating cash flow **EUR 12.5 billion**.
- ▶ Returned **EUR 5.0 billion** to shareholders in 2025: EUR 3.1 billion of dividends, EUR 1.9 billion of buybacks.
- ▶ Dividend of **EUR 1.05 per share**: a **yield of about 4.5%**.

Sources: Eni 2025 results, dividend policy; market data Sept. 2026.

Extra 2

# **Indices and dividends**

## Which indices include dividends?

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Gross total return: dividends  
reinvested

Price only

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DAX 40 (Germany)

S&P 500, Dow Jones, Nasdaq  
(US)

Ibovespa (Brazil)

FTSE 100 (UK)

Nikkei 225 (Japan)

CAC 40 (France)

Hang Seng (Hong Kong)

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- ▶ **Price index:** when a company pays a dividend, its share price drops, and the index does not add that cash back. It only measures price changes.
- ▶ **Total return index:** assumes every dividend is reinvested in the index, before tax, so it grows faster over time.
- ▶ Most price indices also have an official total return version, e.g. the S&P 500 TR or the CAC 40 GR.

## The same index comes in three versions

- ▶ **Price index** (price-only or capital index): follows share prices only. When a company pays a dividend its price drops and the index drops with it; the cash is never added back. The headline S&P 500, Dow Jones, FTSE 100, CAC 40 and Nikkei 225 are all price indices.
- ▶ **Gross total return index** (gross return or performance index): every dividend is reinvested in the index on the day it is paid, before any tax. Examples: the DAX 40 as quoted in the news, the S&P 500 TR, the CAC 40 GR.
- ▶ **Net total return index** (net return index): dividends are reinvested after the withholding tax a foreign investor would pay. The usual benchmark for international funds and ETFs. Examples: MSCI World Net, S&P 500 Net.
- ▶ Same companies, same prices in all three. The only difference is what happens to the dividends, and over the long run that difference is large.

Extra 3

# Variance and covariance, other ways

## Alternative expressions of the variance

- ▶ The variance is the average squared distance to the mean. For  $N$  observations  $x_1, \dots, x_N$  with mean  $\bar{x}$ :

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2$$

- ▶ The same number can be written without any distances, as the **mean of the squares minus the square of the mean**:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N x_i^2 - \bar{x}^2 \qquad \text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

- ▶ This is the **König-Huygens formula**. It is handy with a calculator: you add the values, add their squares, and never compute a single distance.

## König-Huygens step by step

- Start from the definition and expand the square inside the sum:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 = \frac{1}{N} \sum_{i=1}^N (x_i^2 - 2 x_i \bar{x} + \bar{x}^2)$$

- Average each of the three pieces separately:

$$\sigma^2 = \underbrace{\frac{1}{N} \sum_{i=1}^N x_i^2}_{\text{mean of the squares}} - \underbrace{\frac{1}{N} \sum_{i=1}^N 2 x_i \bar{x}}_{\text{middle term}} + \underbrace{\frac{1}{N} \sum_{i=1}^N \bar{x}^2}_{\text{last term}}$$

## König-Huygens step by step (con't)

- Develop the middle and last terms:

$$\frac{1}{N} \sum_{i=1}^N 2 x_i \bar{x} = 2\bar{x} \cdot \frac{1}{N} \sum_{i=1}^N x_i = 2\bar{x} \cdot \bar{x} = 2\bar{x}^2$$

$$\frac{1}{N} \sum_{i=1}^N \bar{x}^2 = \frac{1}{N} \cdot N\bar{x}^2 = \bar{x}^2$$

- Add them up:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N x_i^2 - 2\bar{x}^2 + \bar{x}^2 = \frac{1}{N} \sum_{i=1}^N x_i^2 - \bar{x}^2$$

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

## From data to expectations

- ▶ For data, “average” means  $\frac{1}{N} \sum$ : every observation counts  $1/N$ . For a random variable, “average” means the expectation  $\mathbb{E}[\cdot]$ , where every outcome counts with its probability. Same idea, different weights.
- ▶ Replace each  $\frac{1}{N} \sum$  by  $\mathbb{E}$  and  $\bar{x}$  by  $\mathbb{E}[X]$ :

$$\frac{1}{N} \sum_{i=1}^N x_i^2 - \bar{x}^2 \quad \longrightarrow \quad \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \text{Var}(X)$$

- ▶ **Check with numbers.** Stock A from Lecture 2 had returns 12, -4, 10, 2, with mean 5. Mean of the squares:  $(144 + 16 + 100 + 4)/4 = 66$ . Then  $66 - 5^2 = 41$ . The direct route gave  $(49 + 81 + 25 + 9)/4 = 41$ . Same number.
- ▶ In the lecture we divide by  $T - 1$  instead of  $N$  for historical returns. The identity still works; only the constant in front changes.

## Alternative expressions of the covariance

- **The shortcut formula.** The same trick as König-Huygens works for the covariance: the **mean of the products minus the product of the means**, in expectation and for  $N$  observations  $(x_i, y_i)$  with means  $\bar{x}$ ,  $\bar{y}$ :

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y]$$

$$\text{Cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N x_i y_i - \bar{x} \bar{y}$$

- **Covariance from variances.** Take the variance of  $X + Y$ , remove the two individual variances, and halve what is left:

$$\text{Cov}(X, Y) = \frac{1}{2} [\text{Var}(X + Y) - \text{Var}(X) - \text{Var}(Y)]$$

This is the two-stock portfolio variance formula with weights of 1, read backwards.

Extra 4

# The Markowitz problem

## Markowitz: the optimization problem

- ▶ We have  $N$  stocks and must choose the weights  $x_1, \dots, x_N$ . Write  $\sigma_{ij} = \text{Cov}(R_i, R_j)$  for the covariance between stocks  $i$  and  $j$ .
- ▶ **Target:** the smallest possible portfolio variance.

$$\min_{x_1, \dots, x_N} \text{Var}(R_p) = \sum_{i=1}^N \sum_{j=1}^N x_i x_j \sigma_{ij}$$

- ▶ **Constraints:** reach a chosen expected return  $\mu$ , and invest all the money.

$$\sum_{i=1}^N x_i E[R_i] = \mu \qquad \sum_{i=1}^N x_i = 1$$

- ▶ Adding  $x_i \geq 0$  forbids short sales. Solving once for each  $\mu$  traces out the efficient frontier.

## Markowitz: first-order conditions

- ▶ Attach a multiplier to each constraint,  $\lambda$  for the return and  $\gamma$  for the budget, and write the Lagrangian (the  $\frac{1}{2}$  only cleans up the derivative):

$$\mathcal{L} = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N x_i x_j \sigma_{ij} - \lambda \left( \sum_{i=1}^N x_i E[R_i] - \mu \right) - \gamma \left( \sum_{i=1}^N x_i - 1 \right)$$

- ▶ Partial derivative with respect to each weight  $x_k$ , set to zero: one equation per stock.

$$\frac{\partial \mathcal{L}}{\partial x_k} = \sum_{j=1}^N \sigma_{kj} x_j - \lambda E[R_k] - \gamma = 0$$

that is  $\text{Cov}(R_k, R_p) = \lambda E[R_k] + \gamma$

- ▶ The derivatives with respect to  $\lambda$  and  $\gamma$  give back the two constraints:  $N + 2$  linear equations for  $N + 2$  unknowns.

## Markowitz: the solution

- Stack the weights in a column  $\mathbf{x}$ , the expected returns in a column  $\mu$ , a column of ones  $\mathbf{1}$ , and the covariances in the matrix  $\Sigma$ . The  $N$  first-order conditions become one line:

$$\Sigma \mathbf{x} = \lambda \mu + \gamma \mathbf{1} \quad \implies \quad \mathbf{x}^* = \Sigma^{-1} (\lambda \mu + \gamma \mathbf{1})$$

- $\lambda$  and  $\gamma$  are fixed by the two constraints. With the shorthand  $A = \mathbf{1}'\Sigma^{-1}\mathbf{1}$ ,  $B = \mathbf{1}'\Sigma^{-1}\mu$ ,  $C = \mu'\Sigma^{-1}\mu$ ,  $D = AC - B^2$ :

$$\lambda = \frac{A\mu - B}{D} \qquad \gamma = \frac{C - B\mu}{D}$$

- In practice: one matrix inverse (MINVERSE in Excel) and every optimal portfolio follows.

## Markowitz: the frontier

- Plug the solution back into the variance. The smallest variance reachable for a target return  $\mu$  is

$$\sigma_p^2(\mu) = \frac{A\mu^2 - 2B\mu + C}{D}$$

- A parabola in  $\mu$ : that is the minimum-variance frontier, and its upper half is the efficient frontier. Its lowest point, the global minimum-variance portfolio, has weights  $\mathbf{x} = \Sigma^{-1}\mathbf{1}/A$ .
- **Reading the first-order condition:** in an optimal portfolio, each stock's contribution to risk, its covariance with the portfolio, is a straight-line function of its expected return. Lecture 4 builds the CAPM on exactly this idea.

## Markowitz in real life: no closed form

- ▶ Asset managers solve the same kind of problem every day, but with extra constraints that a real mandate imposes:
  - ▶ **Caps on the weights:** no more than, say, 5% in any single stock, or 25% in any sector.
  - ▶ **No short sales,** or a limit on the total short position.
  - ▶ **Factor neutrality:** the portfolio's exposure to a risk factor (the market, oil, interest rates) must be zero or within a band.
  - ▶ **Turnover and cost limits:** do not trade more than a given fraction of the portfolio.
- ▶ With inequality constraints like these there is no formula to plug into. Instead an **optimizer** searches for the weights numerically, and the result is an approximation, accurate to as many decimals as we ask for.

## Markowitz in real life: an optimizer

- ▶ An open-source example in Python: **CVXPY**. The problem is written almost as it appears on the previous slides: the target to minimise, then the list of constraints, then `solve()`.
- ▶ Here with a return target, full investment, no short sales, and a 5% cap on every weight:

```
import cvxpy as cp
x = cp.Variable(N)
constraints = [mu @ x >= target,      # return target
               cp.sum(x) == 1,        # fully invested
               x >= 0,                 # no short sales
               x <= 0.05]              # 5% cap per stock
objective = cp.Minimize(cp.quad_form(x, Sigma))
cp.Problem(objective, constraints).solve()
```

- ▶ The optimizer returns the weights in `x.value`. Change a constraint, run it again: that is how a real portfolio is rebalanced.